**CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT**

Course Code and Title**: S-CSPC316-Architecture and Organization**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

**DIRECTIONS:** Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

| Format, parts, and description | Compliance | |
|---|---|---|
| | **Complied With** | **Not Complied With** |
| **Format** | | |
| 1. The 2023-2024 syllabus format is used. | / | |
| 2. The font used in the syllabus preparation is Times New Roman (11). | / | |
| 3. DLSUD (being first) and college logos are beside each other at the top (center). | / | |
| 4. Preliminaries include all the contact details of the professor, such as the day and time. | / | |
| 5. Each WEEK has a corresponding date. | / | |
| **Content** | | |
| 6. The Learning Outcomes have more HOTS than LOTS. | / | |
| 7. CLOs and TLOs are identified for each module/lesson/topic. | / | |
| 8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities. | / | |
| 9. The journal articles/references are within the last five years. | / | |

| | | |
|---|---|---|
| 10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies. | / | |
| 11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students. | / | |
| **Assessments** | | |
| 12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus. | / | |
| 13. Formative Assessments are not reflected anymore in the syllabus. | / | |
| 14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities. | / | |
| 15. The syllabus presents one-module(lesson)-one-assessment correspondence. | / | |
| 16. Only one Enabling Assessment is reflected in a given week | / | |
| 17. The Enabling Assessments are aligned with the identified CLOs and TLOs. | / | |
| 18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term. | / | |
| 19. Thirty percent (30%) is allotted for Summative Assessments per term. | / | |
| 20. Fifteen percent 20%) is allotted for Class Participation per term. | / | |
| 21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade. | / | |
| 22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities. | / | |
| 23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities. | / | |
| 24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD | / | |
| 25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD. | / | |
| **References and Curriculum Map** | | |
| 26. The References must be within the last 5 years except for canons and seminal studies. | / | |
| 27. It must be a combination of books, eBooks, journal articles, videos, etc. | / | |

| | | |
|---|---|---|
| 28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS. | / | |
| 29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs) | / | |

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

**Self-reviewed:**



**Jennylinde R. Manaois**

Faculty

**Checked, verified, and endorsed:**



***Rolando B. Barrameda***

Dept. Curriculum Committee Reviewer

**Checked, verified, and approved:**



***Josephine T. Eduardo***

Department Chair



***Dr. Rubie M. Causaren***

College Dean

**DE LA SALLE UNIVERSITY-DASMARINAS**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER SCIENCE DEPARTMENT**

## COURSE SYLLABUS

| | | |
|---|---|---|
| **COURSE CODE** | **:** | S-CSPC316 |
| **COURSE TITLE** | **:** | Architecture and Organization |
| **COURSE TYPE** | **:** | Lecture |
| **COURSE CREDIT** | **:** | 3 Units |
| **MODALITY** | **:** | Traditional blended learning model |
| **PRE-REQUISITES** | **:** | S-ITCS214LA (Data Structures and Algorithm) |
| **CO-REQUISITES** | **:** | None |
| **PROFESSOR** | : | Jennylinde R. Manaois |
| **CONSULTATION HOURS** | : | M / 2:00 -4:00, T / 1:00 -4:00, TH 9:00 – 12:00, 2:00 – 4:00<br>F / 1:00 – 2:00<br>jrmanaois@dlsud.edu.ph |

**COURSE DESCRIPTION:**

This course deals with the introduction to digital hardware design. Combinational and sequential logic circuits are treated. Designing circuits using encoders, decoders, multiplexers, and different types of flip-flops is done for the first part of the course; and then algorithmic state machines are discussed. It also includes concepts relating to computer organization and architecture; instruction set and assembly language; hardware system, central processing unit, and its control; and input-output system and system memory.

**COURSE LEARNING OUTCOMES:**

By the end of this course, students can:

CLO1. distinguish the building blocks that forms computer organization and have knowledge of how to instruct a computer;

CLO2. identify the fundamental circuit components and techniques for optimizing circuits;

CLO3. use Boolean algebra as related to designing computer logic, including solving Karnaugh maps;

CLO4. evaluate the importance of various theories and principles in solving real-life problems by inculcating the usefulness of computer organization;

CLO5. assess computer organization, structure, characteristics, basic theories of the CPU, memory, and I/O peripherals, the operation of modern day computer systems and the fundamental principles that drive computer designer in designing new architectures; and

CLO6. Demonstrate 21st century skills in all learning activities.

**LEARNING PLAN :**

**Learning Outcomes for Midterm Period**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO1. Distinguish the building blocks that forms computer organization and have knowledge of how to instruct a computer. | • TLO1. Understand the main components of a computer.<br>• TLO2. Recognize different types of number systems as they relate to computers.<br>• TLO3. Convert values from decimal, binary, octal, hexadecimal, and binary-coded decimal numbersystems to each other and back to the other systems. |
| CLO2. Identify the fundamental circuit components and techniques for optimizing circuits. | • TLO5. Understand the basic operations of Boolean algebra and construct simple logic circuits.<br>• TLO7. Understand, analyze and design various combinational and sequential circuits. |
| CLO3. Use Boolean algebra as related to designing computer logic, including solving Karnaugh maps. | • TLO6. Use a Karnaugh map to simplify a Boolean expression. |
| CLO4. Evaluate the importance of various theories and principles in solving real-life problems by inculcating the usefulness of computer organization. | • TLO4. Understand the distinction between the way in which numbers are represented (the binary format) and the algorithms used for the basic arithmetic operations. |
| CLO6. Demonstrate 21st century skills in all learning activities. | • Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments) |

| **Module No and Learning Outcomes** | **Date** | **Teaching-Learning Activities / Assessment Strategy** | **Technology Enabler** | **Onsite** | **Asynchronous** | **Allotted Hours** |
|---|---|---|---|---|---|---|
| Module 1,2<br>TLO 1<br>CLO 1,6 | Week 1<br>(Aug.14-19) | • Gospel reading<br>• Syllabus<br>• Presentation of course introduction<br>• Main Components of Computer | Schoolbook, Powerpoint, | / | | 3.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Module 2<br>TLO 1<br>CLO1 | Week 2<br>(Aug. 21-Aug. 26) | • Gospel reading<br>• Main Components of Computer<br>• Enabling assessment: Quiz on Components of Computer | Schoolbook, Powerpoint, Youtube Videos | | / | 3.0 |
| Module 3<br>TLO 2<br>TLO 3<br>TLO 4<br>CLO 1<br>CLO 4 | Week 3<br>(Aug. 28-Sept. 2) | • Gospel reading<br>• Data representations in Computer Systems<br>• Enabling assessment: exercises on Data Representations | Schoolbook, Powerpoint | / | | 3.0 |
| Module 4<br>TLO 5<br>CLO 2 | Week 4<br>(Sept. 4-9) | • Gospel reading<br>• Boolean Algebra and Digital Logic<br>• Enabling assessment: exercises on Boolean Algebra and Data Logic | Schoolbook, Powerpoint, Youtube Videos | | / | 3.0 |
| Module 2,3 | Week 5<br>(Sept. 11-16) | • Review / Consultation of Module 2, 3<br>• Long Quiz 1: Module 2,3 | Schoolbook / MS Teams Schoolbook | / | | 3.0 |
| Module 5<br>TLO 6<br>CLO 3 | Week 6<br>(Sept. 18-23) | • Gospel reading<br>• Karnaugh Maps | Schoolbook, Powerpoint, Youtube Videos | | / | 3.0 |
| Module 5<br>TLO 6<br>CLO 3 | Week 7<br>(Sept. 25-30) | • Gospel reading<br>• Karnaugh Maps<br>• Enabling assessment: quiz on Karnaugh Maps | Schoolbook, Powerpoint, | / | | 3.0 |
| Module 6<br>TLO 7<br>CLO 2 | Week 8<br>(Oct. 2-7) | • Combinational Logic Circuit and Sequential Logic Circuit | Schoolbook, Powerpoint, Youtube Videos | | / | 3.0 |
| Module 4,5,6<br>TLO 5,6,7<br>CLO 2,3, 6 | Week 9<br>(Oct. 9-14) | • Midterm Examination | Schoolbook | / | | 3 |
| | | | ***TOTAL*** | ***15*** | ***12*** | ***27.0*** |

**Learning Outcomes for Final Period**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO1. Distinguish the building blocks that forms computer organization and have knowledge of how to instruct a computer. | • TLO8. Demonstrate knowledge on the different parts of the CPU and can explain how the parts are interconnected with each other.<br>• TLO9. Describe the basic concepts of bus, clocks and I/O subsystem. |

| | |
|---|---|
| CLO4. Evaluate the importance of various theories and principles in solving real-life problems by inculcating the usefulness of computer organization. | • TLO11. Describe the types of operands used in typical machine instruction sets.<br>• TLO12. Demonstrate skill in assembly language. |
| CLO5. Assess computer organization, structure, characteristics, basic theories of the CPU, memory, and I/O peripherals, the operation of modern day computer systems and the fundamental principles that drive computer designer in designing new architectures. | • TLO10. Present an overview of the memory organization and addressing. |
| CLO6. Demonstrate 21st century skills in all learning activities. | • Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments) |

| **Module No and Learning Outcomes** | **Date** | **Teaching-Learning Activities / Assessment Strategy** | **Technology Enabler** | **Onsite** | **Asynchronous** | **Allotted Hours** |
|---|---|---|---|---|---|---|
| | Week 10 (Oct. 16-21) | • Completion of missed assessments | Schoolbook | | / | 3.0 |
| | Week 11 (Oct. 23-28) | Self-care Week | | | | 3.0 |
| Module 7,8<br>TLO 8,9<br>CLO 1 | Week 12 (Oct. 30 – Nov. 4) | • Gospel reading<br>• CPU Basics and Organization<br>• The Bus, Clocks, I/O Subsystem | Schoolbook, Powerpoint, Youtube Videos | | / | 3.0 |
| Module 9<br>TLO 10<br>CLO 5 | Week 13 (Nov. 6 – 11) | • Memory Organization and Addressing<br>• Enabling assessment: quiz on Module 7,8,9 | Schoolbook , Powerpoint | / | | 3.0 |
| Module 10<br>TLO 11<br>CLO 4 | Week 14 (Nov. 13 – 18) | • Gospel reading<br>• Instruction Set Architecture<br>• Enabling assessment: quiz on Instruction Set Architecture | Schoolbook<br>Powerpoint | / | | 3.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Module 7,8,9,10 TLO 8,9,10,11 | Week 15 (Nov. 20 – 25) | • Review / Consultation of Module 7, 8,9<br>• Long Quiz : Module 7, 8, 9,10 | Schoolbook | / | | 3.0 |
| Module 11<br><br>TLO 12 CLO 4 | Week 16 (Nov.27 – Dec. 2) | • Gospel reading<br>• Overview of Assembly Language<br>• Parts of an Assembly Language Program<br>• Enabling Assessment: Assembly Language | Schoolbook , 8086 Emulator | | / | 3.0 |
| Module 11 TLO 12 CLO 4 | Week 17 (Dec. 4-9) | • Understanding an Assembly Language Program<br>• Enabling Assessment: Assembly Language | Schoolbook , 8086 Emulator | / | | 3.0 |
| Module 10, 11 TLO 11, 12, CLO 4 | Week 18 (Dec 11-16) | • Final Examination | Schoolbook | / | | 3.0 |
| | | | ***TOTAL*** | ***15*** | ***12*** | ***27.0*** |

**GRADING SYSTEM :**

**Midterm**

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessment - 30%
100%

**Finals**

Enabling Assessments - 50%
Attendance/Class Participation - 20%
Summative Assessments - 30%
100%

**Overall Final Grade =** $\frac{\text{Midterm + Final}}{2}$

**COURSE POLICIES AND REQUIREMENTS:**

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback

a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
    i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
    ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
    iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

**REFERENCES :**

**Online References**

| E-provider | References |
|---|---|
| https://www.geeksforgeeks.org/computer-organization-and-architecture-tutorials/ | Computer Organization and Architecture Tutorials |
| http://staff.ustc.edu.cn/~llxx/cod/reference_books/Computer%20Organization%20and%20Architecture%209th.pdf | Stallings, William. (2016). Computer Organization and Architecture. Pearson Education, Inc., publishing as Prentice Hall. |
| http://e4uhu.com/down/Logic/5e.pdf | Mano, Morris M.,Ciletti, Michael D. (2013). Digital Design. Pearson Education, Inc., publishing as Prentice Hall, One Lake Street, Upper Saddle River, New Jersey 07458 |
| https://www.youtube.com/watch?list=PL8dPuuaLjXtNlUrzyH5r6jN9ulIgZBpdo&time_continue=10&v=FZGugFqdr60&feature=emb_logo | The Central Processing Unit (CPU): Crash Course Computer Science #7 |
| https://www.youtube.com/watch?v=zltgXvg6r3k&list=PL8dPuuaLjXtNlUrzyH5r6jN9ulIgZBpdo&index=9 | Instructions & Programs: Crash Course Computer Science #8 |

**Onsite References:**

| Provider | References |
|---|---|
| AEA Lib | Null L, Lobur J. (2019). The essentials of computer organization and architecture. Burlington, Massachusetts : Jones & Bartlett Learning. |
| AEA Lib | David A . Patterson ; John L. Hennessy. (2018). Computer organization. Cambridge, Massachusetts : Morgan Kaufmann Publishers, an imprint of Elsevier |
| AEA Lib | Stallings W. (2016). Computer organization and architecture: designing for performance. Boston : Pearson-Prentice Hall. |
| AEA Lib | Clements A. (2014). Computer organization and architecture: themes and variations. Stamford, CT : Cengage Learning. |
| AEA Lib | Shiva, Sajjan G. (2014). **Computer** organization, **design**, and architecture. Boca Raton : CRC Press, Taylor & Francis Group. |

**Prepared:** **Computer Studies Department**
1st Semester 2023-2024

**Approved:** **Josephine T. Eduardo**
Chair, CSD

**Endorsed:** **Rubie M. Causaren, PhD**
Dean, CSCS

**DE LA SALLE UNIVERSITY - DASMARIÑAS**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER SCIENCE DEPARTMENT**

**PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE**

| | | |
|---|---|---|
| **COURSE CODE** | **:** | S-CSPC316 |
| **COURSE TITLE** | **:** | Architecture and Organization |
| **COURSE TYPE** | **:** | Lecture |
| **COURSE CREDIT** | **:** | 3 Units |
| **MODALITY** | **:** | Traditional blended learning model |
| **PREREQUISITE** | **:** | None |

| Course Learning Outcome | Program Learning Outcomes | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 | PLO 8 |
| 1. Distinguish the building blocks that forms computer organization and have knowledge of how to instruct a computer. | ✔ | ✔ | ✔ | | | | ✔ | |
| 2. Identify the fundamental circuit components and techniques for optimizing circuits. | ✔ | ✔ | ✔ | | | | | |
| 3. Use Boolean algebra as related to designing computer logic, including solving Karnaugh maps. | ✔ | ✔ | ✔ | ✔ | | | | |
| 4. Evaluate the importance of various theories and principles in solving real-life problems by inculcating the usefulness of computer organization. | ✔ | ✔ | ✔ | ✔ | ✔ | | | |
| 5. Assess computer organization, structure, characteristics, basic theories of the CPU, memory, and I/O peripherals, the operation of modern day computer systems and the fundamental principles that drive computer designer in designing new architectures. | ✔ | ✔ | ✔ | ✔ | ✔ | | | |
| 6. Demonstrate 21st-century skills in all learning activities. | | | | | ✔ | ✔ | ✔ | ✔ |

*NOTE: Provide a checkmark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)*

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**COMPUTER SCIENCE DEPARTMENT**

**GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME**

**MAPPING TABLE FOR BS COMPUTER SCIENCE**

| Program Learning Outcomes | Graduate Attributes/ Institutional Learning Outcomes | | | | |
|---|---|---|---|---|---|
| | GA1 | GA2 | GA3 | GA4 | GA5 |
| 1. Apply concepts of computing and mathematics in different domains of computer science. | | | | ✔ | ✔ |
| 2. Identify and analyze problems and define computing requirements appropriate to their solutions. | | | | | ✔ |
| 3. Design, develop, test, and evaluate various solutions using **algorithms** for complex computing problems that take full advantage of the usability of computer resources. | | | ✔ | ✔ | ✔ |
| 4. Document computer-based systems by writing effective reports for risk-assessment impact. | | | | ✔ | ✔ |
| 5. Work effectively as an individual or as a member of a team during software development. | | | ✔ | ✔ | ✔ |
| 6. Communicate effectively with the computing community and non-computer science professionals about computing activities. | | | | ✔ | ✔ |
| 7. Recognize the legal, social, ethical, and professional issues involved in the utilization of computer technology and address these issues. | ✔ | ✔ | ✔ | | |
| 8. Engage in continuing professional development through performing independent or collaborative research as computing professionals. | | | ✔ | ✔ | ✔ |

*NOTE: Provide a checkmark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expects from its graduates.*

# GRADUATES ATTRIBUTES AND INSTITUTIONAL LEARNING OUTCOMES (ILOs)

| Graduate Attribute (GA) | Descriptors (Institutional Learning Outcome)[1] |
|---|---|
| 1. God-centered | a) Creates an environment where the experience of God is lived and shared<br>b) Practices honesty, fairness, truth, and integrity in all aspects of life (personal and professional lives)<br>c) Observes and maintains ethical standards in dealing with the different stakeholders<br>d) Integrates Christian perspectives and values in all undertakings<br>e) Manifests humility and respect in relating with other people |
| 2. Patriotic | a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives<br>b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts<br>c) Engages actively in political, social, economic, and cultural transformation for nation building<br>d) Brings pride and honor to the community and the country<br>e) Patronizes locally produced products and promotes them globally |
| 3. Socially Responsive | a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation<br>b) Engages in ecological advocacies as responsible stewards of God's creation<br>c) Continuously works in solidarity with people and institutions to effect liberating action<br>d) Brings forth awareness and deeper understanding of social realities<br>e) Manifests spirit of volunteerism through social actions that improve communities |
| 4. Globally Competitive | a) Takes progressive responsibility for own learning and development<br>b) Promotes critical and creative thinking, self-knowledge and self-mastery<br>c) Is committed to rise above mediocrity<br>d) Shows dedication, commitment and love for work to attain exceptional results<br>e) Exhibits innovativeness and creativity in various contexts |

[1] *Approved in July 2017 during a multi-sectoral committee specifically convened for the purpose of coming up with descriptions for the graduate attributes.*